

Data Tables — Consolidated Appendix

Custom API Integration · SAP S/4HANA → Zetheta FinSight

Project Code: 493560B

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Field Mappings · Data Quality Rules · Error Registry · Test Traceability · Risk Register · Dashboard Panels

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D3 – Field Mapping Specification (85 mappings across 10 domains)

MappingID	Domain	SourceTable	SourceField	TargetField	TransformationRule	ValidationRule	ErrorCode	BusinessContext
MAP-GL-001	General Ledger	ACDOCA	RBUKRS+GJAHR+BELNR	journalEntry.documentId	CONCAT(RBUKRS,'-',GJAHR,'-',LTRIM(BELNR,'0'))	NOT NULL; regex ^[A-Z0-9]{2,6}-\d{4}-\d{1,10}\$	ERR-MAP-001;ERR-VAL-001	Composite business key linking SAP document to FinSight record; idempotency and audit lineage
MAP-GL-002	General Ledger	ACDOCA	BUDAT	journalEntry.postingDate	FORMAT(BUDAT,'YYYY-MM-DD')	NOT NULL; within fiscal year +/-1 month	ERR-MAP-002;ERR-VAL-005	Posting date determines fiscal-period bucket; ISO 8601 required
MAP-GL-003	General Ledger	ACDOCA	BLDAT	journalEntry.documentDate	FORMAT(BLDAT,'YYYY-MM-DD')	NOT NULL; <= postingDate + 30 days	ERR-MAP-002;ERR-VAL-005	Original document date for invoice-to-posting lag analysis
MAP-GL-004	General Ledger	ACDOCA	RACCT	journalEntry.glAccount	LTRIM(RACCT,'0') then lookup analytics CoA mapping	Must exist in target chart of accounts	ERR-MAP-001;ERR-VAL-004	Maps SAP CoA to FinSight analytical CoA; consistency 100%
MAP-GL-005	General Ledger	ACDOCA	HSL	journalEntry.amountLC	DECIMAL(HSL,2) in group currency (INR)	NOT NULL; range -999999999.99 to 999999999.99	ERR-MAP-001;ERR-VAL-002	Local/group-currency amount; feeds debit-credit reconciliation
MAP-GL-006	General Ledger	ACDOCA	WSL	journalEntry.amountTC	DECIMAL(WSL,2) in transaction currency	NOT NULL; same range as amountLC	ERR-MAP-001;ERR-VAL-002	Transaction-currency amount for multi-currency reporting
MAP-GL-007	General Ledger	ACDOCA	RHCUR	journalEntry.localCurrency	Direct map; validate ISO 4217	Valid 3-letter ISO 4217 code	ERR-MAP-004;ERR-VAL-003	Group currency (INR) code; guards RS->INR typo class
MAP-GL-008	General Ledger	ACDOCA	RWCUR	journalEntry.transactionCurrency	Direct map; validate ISO 4217	Valid 3-letter ISO 4217 code	ERR-MAP-004;ERR-VAL-003	Transaction currency; input to point-in-time TCURR conversion
MAP-GL-009	General Ledger	ACDOCA	KOSTL	journalEntry.costCentre	LTRIM(KOSTL,'0') then validate against CC master	If present must exist in cost-centre dimension	ERR-MAP-003;ERR-VAL-004	GL->CC referential link; master syncs before transactional
MAP-GL-010	General Ledger	ACDOCA	PRCTR	journalEntry.profitCentre	LTRIM(PRCTR,'0') then validate against PC master	If present must exist in profit-centre dimension	ERR-MAP-003;ERR-VAL-004	GL->PC referential link; drives segment profitability
MAP-GL-011	General Ledger	ACDOCA	MONAT	journalEntry.fiscalPeriod	MAP 001->April..012->March (V3); 013-016->012 with specialPeriod=true	Range 001-016; flag special periods	ERR-MAP-002;ERR-VAL-002	Fiscal-to-calendar period mapping for India V3; year-end adjustments
MAP-GL-012	General Ledger	ACDOCA/BKPF	BLART+STBLG	journalEntry.documentType	MAP BLART to analytics enum; if STBLG set isReversal=true	Must map to valid analytics document type	ERR-MAP-001;ERR-VAL-003	Document type drives processing rules; reversal flag prevents double count
MAP-GL-013	General Ledger	ACDOCA	RBUKRS	journalEntry.tenantId	MAP MC01->MERIDIAN-MC01 MC02->MERIDIAN-MC02 MC03->MERIDIAN-MC03	Must resolve to one of 3 canonical tenants	ERR-MAP-001;ERR-VAL-003	Canonical multi-tenant routing; selects Kafka partition
MAP-GL-014	General Ledger	ACDOCA	GJAHR	journalEntry.fiscalYear	Direct map; cast integer YYYY	Range 2000-2099; equals year of documentId	ERR-VAL-002	Fiscal year for period-over-period analytics
MAP-GL-015	General Ledger	ACDOCA	BUZEI	journalEntry.lineNumber	CAST(BUZEI AS INT)	NOT NULL; >0; unique within documentId	ERR-MAP-001;ERR-VAL-002	Line-item sequence; guarantees uniqueness of each line
MAP-GL-016	General Ledger	ACDOCA/BSEG	HSL/SHKZG	journalEntry.debitCreditIndicator	IF SHKZG='S' OR HSL>=0 THEN 'DEBIT' ELSE 'CREDIT'	Enum {DEBIT CREDIT}	ERR-MAP-006;ERR-VAL-003	Explicit D/C flag underpins zero-variance debit=credit reconciliation
MAP-GL-017	General Ledger	BKPF	USNAM	journalEntry.postedBy	Direct map; UPPERCASE trim	Max length 12; NOT NULL for posted docs	ERR-MAP-001;ERR-VAL-001	User who posted document; segregation-of-duties audit
MAP-GL-018	General Ledger	BKPF	TCODE	journalEntry.sourceTransactionCode	Direct map; default UNKNOWN if null	Max length 20	ERR-MAP-001	Originating SAP transaction; classifies posting channel
MAP-AP-001	Accounts Payable	BSIK	BUKRS+GJAHR+BELNR+LIFNR	payable.payableId	CONCAT(BUKRS,'-',GJAHR,'-',LTRIM(BELNR,'0'),'-',LTRIM(LIFNR,'0'))	NOT NULL; regex ^[A-Z0-9]{2,6}-\d{4}-\d{1,10}-\d{1,10}\$	ERR-MAP-001;ERR-VAL-001	Composite open-item key; idempotency and lineage
MAP-AP-002	Accounts Payable	BSIK	LIFNR	payable.vendorId	LTRIM(LIFNR,'0') then validate vendor master	Must exist in LFA1	ERR-MAP-001;ERR-VAL-004	Links payable to vendor dimension; master before transaction
MAP-AP-003	Accounts Payable	LFA1	NAME1	payable.vendorName	Enrichment join on LIFNR; TRIM	NOT NULL; max length 35	ERR-MAP-001;ERR-VAL-001	Vendor name for AP ageing and payment reports
MAP-AP-004	Accounts Payable	BSIK	DMBTR+SHKZG	payable.openAmount	DECIMAL(DMBTR,2); sign -DMBTR if SHKZG='S' else +DMBTR	NOT NULL; range -999999999.99 to 999999999.99	ERR-MAP-001;ERR-VAL-002	Open liability amount; core of AP balance and cash-flow forecast
MAP-AP-005	Accounts Payable	LFA1	STCD3/STCD1	payable.gstin	Direct map UPPERCASE; validate GSTIN pattern	GSTIN 15-char regex; PAN regex	ERR-MAP-001;ERR-VAL-001	India GST compliance; invalid GSTINs quarantined not blocking clean records

MappingID	Domain	SourceTable	SourceField	TargetField	TransformationRule	ValidationRule	ErrorCode	BusinessContext
MAP-AP-006	Accounts Payable	BSIK	ZFBDT	payable.ageingBucket	days=DATEDIFF(extractionDate,ZFBDT); bucket 0-30/31-60/61-90/90+	ZFBDT NOT NULL valid date	ERR-MAP-002;ERR-VAL-002	Ageing analysis for AP dashboard and DPO
MAP-AP-007	Accounts Payable	BSIK/LFC1	ZFBDT+ZTERM	payable.dueDate	dueDate=ZFBDT+termDays(ZTERM) then FORMAT	Must be >= ZFBDT	ERR-MAP-002;ERR-VAL-005	Net due date drives payment scheduling and overdue detection
MAP-AP-008	Accounts Payable	LFC1	AKONT	payable.reconciliationAccount	LTRIM(AKONT,'0') then CoA lookup	Must exist in chart of accounts	ERR-VAL-004	Ties AP sub-ledger to GL reconciliation account
MAP-AP-009	Accounts Payable	LFC1	ZTERM	payable.paymentTerms	MAP ZTERM to enum + termDays	Must map to known payment-term code	ERR-VAL-003	Standardises SAP payment-term keys for analytics
MAP-AP-010	Accounts Payable	BSIK	AUGBL+AUGDT	payable.clearingStatus	IF AUGBL<>' ' THEN CLEARED (clearingDate=AUGDT) ELSE OPEN	Enum {OPEN CLEARED}	ERR-VAL-003	Clearing status determines open-items report inclusion
MAP-AP-011	Accounts Payable	BSIK	UMSKZ	payable.specialIndicator	MAP UMSKZ to enum; default NONE if blank	Must map to known indicator	ERR-VAL-003	Distinguishes special GL transactions (down payments)
MAP-AP-012	Accounts Payable	LFA1	LAND1	payable.vendorCountry	MAP SAP country key to ISO 3166 alpha-2	Valid ISO 3166 code	ERR-VAL-003	Domestic vs import; drives IGST vs CGST/SGST treatment
MAP-AR-001	Accounts Receivable	BSID	BUKRS+GJAHR+BELNR+KUNNR	receivable.receiveableId	CONCAT(BUKRS,'-',GJAHR,'-',LTRIM(BELNR,'0'),'-',LTRIM(KUNNR,'0'))	NOT NULL; regex ^[A-Z0-9]{2,6}-\d{4}-\d{1,10}-\d{1,10}\$	ERR-MAP-001;ERR-VAL-001	Composite open-item key; idempotency and lineage
MAP-AR-002	Accounts Receivable	BSID	KUNNR	receivable.customerId	LTRIM(KUNNR,'0') then validate customer master	Must exist in KNA1	ERR-MAP-001;ERR-VAL-004	Links receivable to customer dimension
MAP-AR-003	Accounts Receivable	KNA1	NAME1	receivable.customerName	Enrichment join on KUNNR; TRIM	NOT NULL; max length 35	ERR-MAP-001;ERR-VAL-001	Customer name for AR ageing and collections
MAP-AR-004	Accounts Receivable	BSID	DMBTR+SHKZG	receivable.openAmount	DECIMAL(DMBTR,2); sign +DMBTR if SHKZG='S' else -DMBTR	NOT NULL; range -999999999.99 to 999999999.99	ERR-MAP-001;ERR-VAL-002	Open receivable amount; core of AR balance and DSO
MAP-AR-005	Accounts Receivable	KNA1	STCD3/STCD1	receivable.gstin	Direct map UPPERCASE; validate GSTIN pattern	GSTIN 15-char regex	ERR-MAP-001;ERR-VAL-001	Customer GST identity for GSTR-1 outward-supply reconciliation
MAP-AR-006	Accounts Receivable	BSID	ZFBDT	receivable.ageingBucket	days=DATEDIFF(extractionDate,ZFBDT); bucket 0-30/31-60/61-90/90+	ZFBDT NOT NULL valid date	ERR-MAP-002;ERR-VAL-002	AR ageing for collections prioritisation and DSO
MAP-AR-007	Accounts Receivable	BSID/KNC1	ZFBDT+ZTERM	receivable.dueDate	dueDate=ZFBDT+termDays(ZTERM) then FORMAT	Must be >= ZFBDT	ERR-MAP-002;ERR-VAL-005	Net due date drives overdue detection and dunning trigger
MAP-AR-008	Accounts Receivable	KNKK	KLIMK	receivable.creditLimit	DECIMAL(KLIMK,2); creditExposurePct=openBalance/KLIMK*100	KLIMK >= 0; guard divide-by-zero	ERR-MAP-006;ERR-VAL-002	Credit-limit integration; flags customers over exposure
MAP-AR-009	Accounts Receivable	BSID	MAHNS	receivable.dunningLevel	MAP MAHNS 0-9 to enum {NONE REMINDER DUNNING_1 DUNNING_2 LEGAL}	Range 0-9	ERR-VAL-002;ERR-VAL-003	Dunning level for collections workflow and bad-debt risk
MAP-AR-010	Accounts Receivable	BSID	AUGBL+AUGDT	receivable.clearingStatus	IF AUGBL<>' ' THEN CLEARED (clearingDate=AUGDT) ELSE OPEN	Enum {OPEN CLEARED}	ERR-VAL-003	Clearing status determines open-items report inclusion
MAP-AR-011	Accounts Receivable	KNC1	AKONT	receivable.reconciliationAccount	LTRIM(AKONT,'0') then CoA lookup	Must exist in chart of accounts	ERR-VAL-004	Ties AR sub-ledger to GL reconciliation account
MAP-AR-012	Accounts Receivable	KNA1	LAND1	receivable.customerCountry	MAP SAP country key to ISO 3166 alpha-2	Valid ISO 3166 code	ERR-VAL-003	Domestic vs export; export zero-rated / IGST treatment
MAP-CC-001	Cost Centre	CSKS	KOSTL	costCentre.costCentreId	LTRIM(KOSTL,'0')	NOT NULL; regex ^[A-Z0-9]{1,10}\$; unique within controlling area	ERR-MAP-001;ERR-VAL-001;ERR-VAL-006	Primary key of cost-centre dimension; target for GL->CC references
MAP-CC-002	Cost Centre	CSKT	KTEXT	costCentre.costCentreName	Enrichment join (KOKRS KOSTL DATBI) SPRAS='E'; TRIM	NOT NULL; max length 20	ERR-MAP-001;ERR-VAL-001	Short name for cost reports and variance analysis
MAP-CC-003	Cost Centre	CSKT	LTEXT	costCentre.description	Enrichment join SPRAS='E'; TRIM	Optional; max length 40	ERR-MAP-001	Long descriptive text for dashboard tooltips
MAP-CC-004	Cost Centre	CSKS	KOKRS	costCentre.controllingArea	Direct map	NOT NULL; exists in controlling-area master	ERR-VAL-003;ERR-VAL-004	Controlling area scopes the cost-centre namespace
MAP-CC-005	Cost Centre	CSKS	BUKRS	costCentre.tenantId	MAP MC01->MERIDIAN-MC01 MC02->MERIDIAN-MC02 MC03->MERIDIAN-MC03	Must resolve to canonical tenant	ERR-MAP-001;ERR-VAL-003	Routes master record to correct FinSight tenant

MappingID	Domain	SourceTable	SourceField	TargetField	TransformationRule	ValidationRule	ErrorCode	BusinessContext
MAP-CC-006	Cost Centre	CSKS	KOSAR	costCentre.category	MAP KOSAR to enum {PRODUCTION SERVICE ADMIN SALES DEVELOPMENT MANAGEMENT}	Must map to known category	ERR-VAL-003	Cost-centre category drives allocation and overhead pooling
MAP-CC-007	Cost Centre	CSKS	VERAK	costCentre.personResponsible	Direct map; TRIM	Optional; max length 20	ERR-MAP-001	Accountability owner for variance follow-up
MAP-CC-008	Cost Centre	CSKS	DATAB+DATBI	costCentre.validFrom	FORMAT(DATAB,'YYYY-MM-DD'); FORMAT(DATBI,'YYYY-MM-DD')	validFrom <= validTo; DATBI default 9999-12-31	ERR-MAP-002;ERR-VAL-005	Time-dependent validity for point-in-time-correct joins
MAP-CC-009	Cost Centre	CSKS/SETNODE/SETLEAF	KHINR+SETNODE+SETLEAF	costCentre.hierarchyLevel1..7	Recursive SETNODE/SETLEAF walk root->leaf KOSTL into Level1..Level7; collapse >7 into Level7 null-fill shallower	Leaf resolves to one root path; no cycles	ERR-MAP-003;ERR-VAL-004	Flattens 7-level hierarchy for dimensional modelling and roll-ups
MAP-PC-001	Profit Centre	CEPC	PRCTR	profitCentre.profitCentreId	LTRIM(PRCTR,'0')	NOT NULL; regex ^[A-Z0-9]{1,10}\$; unique within controlling area	ERR-MAP-001;ERR-VAL-001;ERR-VAL-006	Primary key of profit-centre dimension; target for GL->PC references
MAP-PC-002	Profit Centre	CEPCT	KTEXT	profitCentre.profitCentreName	Enrichment join (PRCTR DATBI) SPRAS='E'; TRIM	NOT NULL; max length 20	ERR-MAP-001;ERR-VAL-001	Name for profitability and segment reports
MAP-PC-003	Profit Centre	CEPC	SEGMENT	profitCentre.segment	Direct map; validate against segment master	Must exist in segment master; NOT NULL for reporting PCs	ERR-MAP-003;ERR-VAL-004	Segment underpins IFRS 8 / Ind AS 108 segment reporting
MAP-PC-004	Profit Centre	CEPC	KOKRS	profitCentre.controllingArea	Direct map	NOT NULL; exists in controlling-area master	ERR-VAL-003;ERR-VAL-004	Controlling area scopes the profit-centre namespace
MAP-PC-005	Profit Centre	CEPC/SETNODE/SETLEAF	KHINR+SETNODE+SETLEAF	profitCentre.hierarchyLevel1..7	Recursive SETNODE/SETLEAF walk root->leaf PRCTR into Level1..Level7; collapse >7 null-fill shallower	Leaf resolves to one root path; no cycles	ERR-MAP-003;ERR-VAL-004	Flattens hierarchy for segment roll-ups and drill-down
MAP-PC-006	Profit Centre	CEPC	DATAB+DATBI	profitCentre.validFrom	FORMAT(DATAB,'YYYY-MM-DD'); FORMAT(DATBI,'YYYY-MM-DD')	validFrom <= validTo; DATBI default 9999-12-31	ERR-MAP-002;ERR-VAL-005	Time-dependent validity for point-in-time-correct joins
MAP-ML-001	Material Ledger	MBEW	MATNR+BWKEY	materialCost.materialCostId	CONCAT(LTRIM(MATNR,'0'),'-',BWKEY)	NOT NULL; unique per material+valuation area	ERR-MAP-001;ERR-VAL-001;ERR-VAL-006	Composite key of the material-valuation record
MAP-ML-002	Material Ledger	MBEW	BWKEY	materialCost.valuationArea	Direct map; enrich plant->company code	NOT NULL; exists in plant master	ERR-VAL-004	Valuation area scopes the material cost; links to MC0x tenant
MAP-ML-003	Material Ledger	MBEW	VPRSV	materialCost.priceControl	MAP VPRSV S->STANDARD V->MOVING_AVERAGE	Must be S or V	ERR-VAL-003	Price-control indicator governs authoritative price
MAP-ML-004	Material Ledger	MBEW	STPRS+PEINH	materialCost.standardPriceUnit	DECIMAL(STPRS,NULLIF(PEINH,0),2)	PEINH != 0; within amount range	ERR-MAP-006;ERR-VAL-002	Unit standard cost for planned-cost variance and valuation
MAP-ML-005	Material Ledger	MBEW	VERPR+PEINH	materialCost.movingAvgPriceUnit	DECIMAL(VERPR,NULLIF(PEINH,0),2)	PEINH != 0; within amount range	ERR-MAP-006;ERR-VAL-002	Actual moving-average cost for actual-costing analytics
MAP-ML-006	Material Ledger	MBEW	SALK3+LBKUM	materialCost.totalStockValue	DECIMAL(SALK3,2); DECIMAL(LBKUM,3)	Range checks; SALK3 sign-consistent	ERR-MAP-001;ERR-VAL-002	Total inventory value/quantity for balance-sheet inventory line
MAP-ML-007	Material Ledger	MBEW/COSP	SALK3+STPRS+LBKUM+PEINH+WKG001..016	materialCost.priceDifference	priceDifference=SALK3-(LBKUM*STPRS/NULLIF(PEINH,0)); actualCost=SALK3+SUM(COSP price-diff cols)	PEINH != 0; components reconcile to SALK3 within tolerance	ERR-MAP-006;ERR-VAL-002	Price-difference allocation reconciles standard to actual cost (ML run)
MAP-PO-001	Purchase Orders	EKPO	EBELN+EBELP	purchaseOrder.poLineId	CONCAT(LTRIM(EBELN,'0'),'-',LTRIM(EBELP,'0'))	NOT NULL; regex ^\d{1,10}-\d{1,5}\$; unique	ERR-MAP-001;ERR-VAL-001;ERR-VAL-006	Composite PO-line key; idempotency and procurement lineage
MAP-PO-002	Purchase Orders	EKKO	LIFNR	purchaseOrder.vendorId	LTRIM(LIFNR,'0') then validate vendor master	Must exist in LFA1	ERR-MAP-001;ERR-VAL-004	Links PO to vendor dimension; spend-by-vendor analytics
MAP-PO-003	Purchase Orders	EKPO	MATNR	purchaseOrder.materialId	LTRIM(MATNR,'0') then validate material master	Must exist (blank allowed for text POs)	ERR-VAL-004	Links PO line to material; procurement-cost analysis
MAP-PO-004	Purchase Orders	EKPO	NETPR+PEINH	purchaseOrder.netPriceUnit	netPriceUnit=DECIMAL(NETPR/NULLIF(PEINH,0),2); lineNetValue=netPriceUnit*MENGE	PEINH != 0; range checks	ERR-MAP-006;ERR-VAL-002	Pricing-condition net price per base unit and extended line value
MAP-PO-005	Purchase Orders	EKPO/EKET	MENGE+WEMNG	purchaseOrder.grisStatus	openGrQty=SUM(MENGE)-SUM(WEMNG); status FULLY_RECEIVED/NOT_RECEIVED/PARTIAL	WEMNG <= MENGE; qty >= 0	ERR-MAP-006;ERR-VAL-002	GR/IR reconciliation; detects unbilled receipts / undelivered POs
MAP-PO-006	Purchase Orders	EKET	EINDT	purchaseOrder.deliveryDate	FORMAT(EINDT,'YYYY-MM-DD') earliest open schedule line	Valid date; >= AEDAT	ERR-MAP-002;ERR-VAL-005	Committed delivery date for OTD and supply-risk metrics

MappingID	Domain	SourceTable	SourceField	TargetField	TransformationRule	ValidationRule	ErrorCode	BusinessContext
MAP-SO-001	Sales Orders	VBAP	VBELN+POSNR	salesOrder.soLineId	CONCAT(LTRIM(VBELN,'0'), '-', LTRIM(POSNR,'0'))	NOT NULL; regex ^\d{1,10}-\d{1,6}\$; unique	ERR-MAP-001;ERR-VAL-001;ERR-VAL-006	Composite SO-line key; order-to-cash lineage
MAP-SO-002	Sales Orders	VBAK	KUNNR	salesOrder.customerId	LTRIM(KUNNR,'0') then validate customer master	Must exist in KNA1	ERR-MAP-001;ERR-VAL-004	Links SO to customer dimension; revenue-by-customer
MAP-SO-003	Sales Orders	VBAP	NETWR+WAERK	salesOrder.netValue	DECIMAL(NETWR,2); WAERK validated ISO 4217	Range checks; WAERK valid ISO 4217	ERR-MAP-001;ERR-MAP-004;ERR-VAL-002	Order net value (transaction currency) for order-book reporting
MAP-SO-004	Sales Orders	VBEP/VBAP	KWMENG+BMENG	salesOrder.revenueRecognitionStage	IF BMENG=0 ORDERED elif BMENG<KWMENG PARTIALLY_CONFIRMED elif delivered DELIVERED else CONFIRMED	Enum {ORDERED CONFIRMED PARTIALLY_CONFIRMED PARTIALLY_DELIVERED DELIVERED}	ERR-MAP-006;ERR-VAL-003	Revenue-recognition stage for order-to-cash progression and forecast
MAP-SO-005	Sales Orders	VBEP	EDATU	salesOrder.requestedDeliveryDate	FORMAT(EDATU,'YYYY-MM-DD') earliest schedule line	Valid date; >= VBAK.ERDAT	ERR-MAP-002;ERR-VAL-005	Requested delivery date for OTIF service metrics
MAP-FA-001	Fixed Assets	ANLA	BUKRS+ANLN1+ANLN2	fixedAsset.assetId	CONCAT(BUKRS,'-',LTRIM(ANLN1,'0'), '-',ANLN2)	NOT NULL; regex ^[A-Z0-9]{2,6}-\d{1,12}-\d{1,4}\$; unique	ERR-MAP-001;ERR-VAL-001;ERR-VAL-006	Composite asset key (main+sub number); asset-register lineage
MAP-FA-002	Fixed Assets	ANLA	ANLKL	fixedAsset.assetClass	MAP ANLKL to enum {LAND BUILDING PLANT_MACHINERY FURNITURE VEHICLE IT_EQUIPMENT INTANGIBLE}	Must map to known asset class	ERR-VAL-003	Asset class governs Schedule II useful-life and depreciation grouping
MAP-FA-003	Fixed Assets	ANLA	TXT50	fixedAsset.description	Direct map; TRIM	NOT NULL; max length 50	ERR-MAP-001;ERR-VAL-001	Asset description for fixed-asset register and audit
MAP-FA-004	Fixed Assets	ANLZ	KOSTL+PRCTR	fixedAsset.costCentre	LTRIM(KOSTL,'0'); LTRIM(PRCTR,'0') then validate masters	If present must exist in CC/PC dimensions	ERR-MAP-003;ERR-VAL-004	Cost/profit-centre assignment drives depreciation posting allocation
MAP-FA-005	Fixed Assets	ANLB/ANLP/ANLA	AFASL+NAFAZ+AKTIV	fixedAsset.depreciationMethod	MAP AFASL to method enum {SLM WDV MANUAL NO_DEP}; periodicDepreciation=DECIMAL(NAFAZ,2); capitalizationDate=FORMAT(AKTIV)	Method maps; NAFAZ in range; AKTIV valid or null (AuC)	ERR-MAP-001;ERR-VAL-002;ERR-VAL-003	Depreciation method + periodic charge for asset roll-forward
MAP-BNK-001	Bank Statements	FEBKO/FEBEP	KUKEY+AZNUM+ESNUM	bankStatement.statementsLineId	CONCAT(KUKEY,'-',AZNUM,'-',LPAD(ESNUM,4,'0'))	NOT NULL; unique per statement line	ERR-MAP-001;ERR-VAL-001;ERR-VAL-006	Composite key of a statement line; idempotency and cash lineage
MAP-BNK-002	Bank Statements	FEBKO	KONTS	bankStatement.bankAccount	Normalise to canonical account/IBAN; strip separators UPPERCASE	NOT NULL; passes checksum where applicable	ERR-MAP-001;ERR-VAL-001	Normalised house-bank account across bank formats
MAP-BNK-003	Bank Statements	FEBEP	KWBTR+VGTYT	bankStatement.lineAmount	DECIMAL(KWBTR,2); direction CREDIT/DEBIT from VGTYT; signed accordingly	Range checks; direction enum {DEBIT CREDIT}	ERR-MAP-001;ERR-VAL-002;ERR-VAL-003	Cash movement amount and direction for cash-position analytics
MAP-BNK-004	Bank Statements	FEBEP	VALUT	bankStatement.valueDate	FORMAT(VALUT,'YYYY-MM-DD')	Valid date	ERR-MAP-002;ERR-VAL-005	Value date drives cash-flow timing and interest reconciliation
MAP-BNK-005	Bank Statements	FEBEP	VBLNR	bankStatement.clearingStatus	IF VBLNR<>' THEN CLEARED (clearingDocument=VBLNR) ELSE UNMATCHED	Enum {CLEARED UNMATCHED}	ERR-VAL-003	Auto-clearing status; unmatched lines to bank-reconciliation exception

D5 — Data Quality Rules (27)

Rule ID	Category	DMBOK dimension	Domain	Rule description	Severity	Action on failure (D4 code)
DQ-001	Null check	Completeness	GL	ACDOCA.RACCT (GL account) → glAccount must be NOT NULL	CRITICAL	ERR-MAP-001 → DLQ
DQ-002	Null check	Completeness	GL	documentId composite (BUKRS-GJAHR-BELNR) must be fully populated	CRITICAL	ERR-MAP-001 → DLQ
DQ-003	Null check	Completeness	GL	postingDate (BUDAT) must be NOT NULL	CRITICAL	ERR-MAP-001 → DLQ
DQ-004	Null check	Completeness	AP	Vendor id (LFA1.LIFNR) must be NOT NULL on payable	CRITICAL	ERR-MAP-001 → DLQ
DQ-005	Null check	Completeness	AR	Customer id (KNA1.KUNNR) must be NOT NULL on receivable	CRITICAL	ERR-MAP-001 → DLQ
DQ-006	Format validation	Validity	GL	postingDate/documentDate conform to ISO 8601 YYYY-MM-DD	HIGH	ERR-MAP-002 / INVALID_DATE_FORMAT → correct once, else DLQ
DQ-007	Format validation	Validity	All	Currency code is valid 3-letter ISO 4217	HIGH	ERR-MAP-004 / INVALID_CURRENCY_CODE → correct once, else DLQ
DQ-008	Format validation	Validity	GL	documentId matches ^[A-Z0-9]{2,6}-\d{4}-\d{1,10}\$	HIGH	ERR-MAP-002 → DLQ
DQ-009	Format validation	Validity	AP/AR	Vendor/customer GSTIN matches 15-char GSTIN regex	HIGH	ERR-MAP-002 → BizEx (notify AP/AR)
DQ-010	Format validation	Validity	Bank	IBAN/account number matches expected bank-format pattern	MEDIUM	ERR-VAL-007 → load with flag
DQ-011	Range check	Validity/Accuracy	GL	amountLC within [-999999999.99, 999999999.99]	HIGH	ERR-MAP-006 → DLQ
DQ-012	Range check	Validity	GL	Fiscal period ∈ 001–012 or special 013–016 (variant V3)	HIGH	ERR-VAL-007 → DLQ
DQ-013	Range check	Validity	GL	postingDate within fiscal year ± 1 month (no future posting)	HIGH	ERR-VAL-007 → DLQ
DQ-014	Range check	Accuracy	ML	Exchange rate > 0 and within ±20% band of prior rate	MEDIUM	ERR-MAP-005 → stale-rate flag
DQ-015	Range check	Validity	FA	Depreciation amount ≤ asset acquisition value	HIGH	ERR-MAP-006 → DLQ
DQ-016	Referential integrity	Consistency	GL→CC	Cost centre exists in cost-centre master before load	CRITICAL	ERR-MAP-003 / REFERENTIAL_INTEGRITY → BizEx
DQ-017	Referential integrity	Consistency	GL→PC	Profit centre exists in profit-centre master	CRITICAL	ERR-RECON-003 → resync master
DQ-018	Referential integrity	Consistency	AP→vendor	Vendor exists in vendor master before payable loads	CRITICAL	ERR-MAP-003 / RESOURCE_NOT_FOUND → BizEx
DQ-019	Referential integrity	Consistency	AR→customer	Customer exists in customer master before receivable loads	CRITICAL	ERR-RECON-003 → resync master
DQ-020	Referential integrity	Consistency	GL	Target glAccount exists in analytics chart-of-accounts mapping	CRITICAL	ERR-MAP-003 → BizEx
DQ-021	Cross-field validation	Accuracy	GL	Per document SUM(debit HSL) = SUM(credit HSL) (double-entry)	CRITICAL	ERR-RECON-002 → quarantine doc, break report
DQ-022	Cross-field validation	Accuracy	AP/AR	Invoice amount = net + CGST + SGST + IGST (GST breakdown preserved)	CRITICAL	ERR-RECON-002 → BizEx
DQ-023	Cross-field validation	Accuracy	GL	documentDate ≤ postingDate ≤ documentDate + 30 days	MEDIUM	ERR-VAL-007 → flag
DQ-024	Cross-field validation	Accuracy	PO	GR quantity ≤ ordered quantity (GR/IR sanity)	HIGH	ERR-MAP-006 → DLQ
DQ-025	Business rule	Uniqueness	All	Business key (documentId) unique in target (no duplicate load)	CRITICAL	DUPLICATE_ENTRY / ERR-LOAD-004 → idempotent skip or alert
DQ-026	Business rule	Validity	All	Company code ∈ {MC01, MC02, MC03} routed to matching tenant	CRITICAL	ERR-MAP-006 → DLQ (mis-route guard)
DQ-027	Business rule	Timeliness	All	Record extracted+loaded within domain SLA window (≤ 4 h freshness)	MEDIUM	ERR-RECON-001 (timeliness flag) → dashboard

D4 – Integration Error Code Registry

Code	Top class	Sub-class	Source	Description	Default action	Retry policy
ERR-EXT-001	TRANSIENT	NETWORK	SAP RFC	RFC connection timeout after 30 s	Retry with exponential backoff + jitter	base 2 s, cap 60 s, max 3
ERR-EXT-002	TRANSIENT	RESOURCE	SAP RFC	RFC connection pool exhausted (all 50 in use)	Wait and retry; fire pool-utilisation alert	fixed 60 s, max 3
ERR-EXT-003	PERMANENT	CONTRACT	SAP ODP	ODP subscription invalidated (provider structure changed)	Alert integration team; halt extraction for this domain	none (0)
ERR-EXT-004	TRANSIENT	NETWORK	SAP ODP	ODP delta returned empty despite known changes	Retry once; if still empty check ODP monitor (RSA1)	base 2 s, max 1
ERR-MAP-001	DATA_QUALITY	DQ-FIELD	Transformation	Source field NULL but target requires NOT NULL	Apply default if defined, else DLQ	1 correction attempt
ERR-MAP-002	DATA_QUALITY	DQ-FIELD	Transformation	Date format invalid (not YYYYMMDD / not a valid date)	Attempt format correction; if fails, DLQ	1 correction attempt
ERR-MAP-003	DATA_QUALITY	DQ-REFERENTIAL	Transformation	Referenced cost centre not found in master data	Route to Business Exception Queue; notify functional team	none (0)
ERR-MAP-004	DATA_QUALITY	DQ-FIELD	Transformation	Currency code not found in ISO 4217 lookup	Check common typos (RS vs INR); else DLQ	1 correction attempt
ERR-MAP-005	DATA_QUALITY	DQ-FIELD	Transformation	Exchange rate not found for currency pair + date	Use nearest available date rate with stale-rate flag	1 correction attempt
ERR-MAP-006	PERMANENT	CONTRACT	Transformation	Transformation rule execution error (div-by-zero, overflow)	Log full context, route to DLQ, alert engineering	none (0)
ERR-VAL-007	DATA_QUALITY	DQ-FIELD	Transformation/Validation	Future posting date (outside fiscal year $\pm$ 1 month)	Route to DLQ; flag for functional review	none (0)
ERR-LOAD-001	TRANSIENT	THROTTLING	FinSight API	HTTP 429 Too Many Requests	Honour Retry-After, backoff + jitter	base 2 s, cap 60 s, max 5
ERR-LOAD-002	TRANSIENT	THROTTLING	FinSight API	HTTP 503 Service Unavailable	Linear backoff 60 s intervals, then circuit break	fixed 60 s, max 5
ERR-LOAD-003	PERMANENT	BUSINESS	FinSight API	HTTP 422 Business Rule Violation	Route to Business Exception Queue, no retry	none (0)
ERR-LOAD-004	PERMANENT	BUSINESS	FinSight API	HTTP 409 Duplicate Entry (idempotent)	Compare payloads: skip if identical, alert if different	none (0)
ERR-LOAD-005	PERMANENT	AUTH	FinSight API	HTTP 401 Authentication Failure	Refresh token once; if still failing, alert Security	1 refresh only
ERR-RECON-001	DATA_QUALITY	DQ-RECON	Reconciliation	Record-count mismatch between source and target	Generate break report, trigger investigation workflow	none (0)
ERR-RECON-002	DATA_QUALITY	DQ-RECON	Reconciliation	Checksum variance exceeds tolerance threshold	Generate break report, escalate to Data Steward	none (0)
ERR-RECON-003	DATA_QUALITY	DQ-RECON	Reconciliation	Referential integrity violation in target	Identify missing master data, trigger master-data resync	none (0)
ERR-SYS-001	SYSTEM	SYSTEM-INFRA	Infrastructure	Kafka broker unreachable	Trigger circuit breaker, alert infrastructure P1	park + replay when CB closes
ERR-SYS-002	SYSTEM	SYSTEM-INFRA	Infrastructure	Disk space below 10% on processing node	Alert infra, archive old logs, request expansion	park + replay

D7 — Test Traceability Matrix (25 scenarios)

Test ID	Verifies (requirement / deliverable)	Domain(s) / endpoint	Error class / code	Recon dimension	SLA / DQ target
TST-FNC-001	Happy-path extract→transform→load (D1/D3)	GL (SRC/DST-001)	—	Completeness, Accuracy	Completeness > 99.5 %, Accuracy > 99.9 %
TST-FNC-002	Currency conversion + ageing + enrichment (D3)	AP (SRC/DST-002)	DATA_QUALITY (ERR-MAP-004/005)	Accuracy	Accuracy > 99.9 %
TST-FNC-003	Master-data delta before transactional (D1 §4)	Cost Centre (SRC/DST-004)	—	Consistency	Freshness ≤ 4 h
TST-FNC-004	Company-code → tenant routing (Canonical §2)	GL, all (MC01/02/03)	—	Consistency	Consistency 100 %
TST-FNC-005	Fiscal period + special-period mapping (D3)	GL (periods 001–016)	—	Validity	Validity > 99 %
TST-FNC-006	Hierarchy flattening L1–L7 (D3)	Cost Centre / Profit Centre	—	Consistency	Consistency 100 %
TST-FNC-007	P2P document-flow linkage (D3)	PO, Material Ledger, AP	—	Consistency	Consistency 100 %
TST-FNC-008	Clearing-status mapping (D3)	Bank Statements (SRC/DST-010)	—	Accuracy	Accuracy > 99.9 %
TST-FNC-009	Budget vs actual variance (D3)	Budget/Actual (SRC/DST-011), Cost Centre	—	Accuracy	Accuracy > 99.9 %
TST-FNC-010	Full zero-break reconciliation (D5)	All 10 domains / 12 EPs	ERR-RECON-*	All 4 dimensions	Zero-break target
TST-NFR-001	Peak throughput within SLA (D1 NFR)	GL	SYSTEM	Timeliness	Batch within SLA > 98 %, ≤ 2 h
TST-NFR-002	Concurrency vs RFC pool ≤ 50 (Canonical §5)	All 12 EPs	TRANSIENT (ERR-EXT-002)	Timeliness	RFC ≤ 50
TST-NFR-003	API latency under load (D1 NFR)	FinSight API	TRANSIENT (ERR-LOAD-001)	Timeliness	P95 < 5 s
TST-NFR-004	Volume scalability / ceiling (D1 NFR)	GL	—	Timeliness	Linear scaling
TST-NFR-005	Endurance / leak-free 24 h (D1 NFR)	All	SYSTEM (ERR-SYS-002)	Timeliness	Stable throughput
TST-FLR-001	Circuit breaker + no-loss recovery (D4)	GL / SAP RFC	TRANSIENT (ERR-EXT-001/002)	Completeness	No data loss
TST-FLR-002	429 backoff w/ jitter (D4, Appendix B)	FinSight API	TRANSIENT (ERR-LOAD-001)	Completeness	No data loss
TST-FLR-003	Kafka failover, ISR recovery (D1/D4)	Kafka broker	SYSTEM (ERR-SYS-001)	Completeness	Zero message loss
TST-FLR-004	Malformed → DLQ routing (D4)	Any / DLQ	DATA_QUALITY (ERR-MAP-*)	Validity	Validity > 99 %
TST-FLR-005	Buffering + eventual consistency (D4)	FinSight target	TRANSIENT	Consistency	Eventual consistency
TST-SEC-001	OAuth auto-refresh (D2/D4, Appendix B)	FinSight auth	PERMANENT→recover (TOKEN_EXPIRED)	—	No data loss
TST-SEC-002	Unauthorised access → 401 + audit (D2)	FinSight auth	PERMANENT (INVALID_TOKEN)	—	Zero data exposure
TST-SEC-003	Encryption in transit/at rest (D1 security)	Kafka, DLQ, transport	SYSTEM	—	TLS 1.2+ / AES-256
TST-REC-001	Variance break detection + alert (D5/D6)	Any (INR 1.50)	DATA_QUALITY (ERR-RECON-002)	Accuracy	Zero unexplained variance
TST-REC-002	Referential-integrity orphan routing (D5)	GL→Cost Centre	DATA_QUALITY (ERR-RECON-003)	Consistency	Consistency 100 %

D1 — Risk Register (12)

ID	Category	Description	Prob.	Impact	Score	Mitigation	Owner
R-01	SAP performance	RFC pool exhaustion — integration exceeds 50 concurrent RFC connections, blocking SAP dialog users	M	H	6	Hard cap connection pool at < 50; circuit breaker on RFC boundary; back-pressure from Kafka; alert at 80% utilisation; co-locate Extractor on-prem to shorten connection hold time	SAP Basis Admin (Priya Deshmukh)
R-02	Schema / data	ODP / CDS schema change — a support pack alters a CDS view (e.g. I_JournalEntry), breaking mappings	M	H	6	Version API contracts; schema-drift detection alert; graceful degradation (route affected records to DLQ, continue clean records); mapping change process	Zetheta Platform Eng (Marcus Wei)
R-03	Network	Bandwidth saturation — integration exceeds 25% of the 450 Mbps MPLS in business hours	M	M	4	Rate-shaping + gzip compression; claim-check offload; schedule bulk/master loads off-peak; monitor WAN utilisation with alert at 20%	VP IT Infrastructure (Rajesh Venkataraman)
R-04	Compliance	Data residency breach — a component, log sink or backup provisioned outside India (RBI violation)	L	H	3	Pin every service, S3 bucket, RDS, and ELK to ap-south-1; org-level SCP denying other regions; automated residency audit; no cross-region replication	VP IT Infrastructure (Rajesh Venkataraman)
R-05	Data integrity	Reconciliation break — source/target counts or amounts diverge, or debit≠credit per document	M	H	6	Automated per-document + per-domain recon; targeted re-load of missing docs from lineage; PagerDuty on break; zero-break gate before sign-off	Head of Internal Audit (Dr. Kulkarni)
R-06	Resilience	DLQ overflow — sustained failures fill the DLQ faster than it is drained	L	M	2	Extended-retention DLQ (topic + S3 overflow); dlq_depth metric with alert; on-call runbook for bulk reprocess; root-cause triage SLA	Zetheta Platform Eng (Marcus Wei)
R-07	Security	Token / auth failure — OAuth token expiry or revocation halts FinSight loads	M	M	4	Automatic refresh-token flow on TOKEN_EXPIRED; re-auth on INVALID_TOKEN with security alert; monitor auth-failure rate; secrets in managed store with rotation	Zetheta Platform Eng (Marcus Wei)
R-08	Availability	Kafka broker loss — a broker fails in ap-south-1	M	M	4	RF=3 / ISR=2 tolerates one broker loss with no data loss; one broker per AZ; automated broker replacement; under-replicated-partition alert	VP IT Infrastructure (Rajesh Venkataraman)
R-09	Scheduling	Batch-window overrun — extraction runs into the 01:00–04:30 IST RGGBS000 window	M	M	4	Scheduler encodes batch + maintenance windows; job time-boxing; early-warning alert if a batch job trends late; defer non-critical loads	SAP Basis Admin (Priya Deshmukh)
R-10	Compliance	GST breakdown loss — CGST/SGST/IGST components dropped during amount transformation	L	H	3	Dedicated GST Preserver component; validation rule asserting components sum to invoice amount; route violations to DLQ; sample audit of transforms	Head of Internal Audit (Dr. Kulkarni)
R-11	Data integrity	Master-data lag — transactional records load before their master data, breaking referential integrity	M	M	4	Enforce master-before-transactional sequencing in Scheduler; referential-integrity validation (GL→CC/PC, AP→vendor, AR→customer); quarantine orphans to DLQ	Zetheta Platform Eng (Marcus Wei)
R-12	Scalability	Festival volume spike — a 5× seasonal surge overwhelms transform/load capacity	M	H	6	Partition for 10× baseline (5× is headroom); autoscale on consumer lag; Kafka log absorbs surge while consumers catch up; bulkhead per domain; load-test the spike scenario	VP IT Infrastructure (Rajesh Venkataraman)

D6 — Monitoring Dashboard Panels (12)

Panel ID	Title	Purpose	Data source	Metric / PromQL query	Visualisation	Refresh	Alert threshold	Severity
MON-001	Pipeline Health Overview	Single at-a-glance verdict across all 10 domains — the first thing on-call reads	Prometheus (recording rule pipeline:health:state)	Status aggregation: GREEN if all domains synced within SLA, YELLOW if any >2 h stale, RED if any >4 h stale or last run failed	Stat / state-timeline (colour-coded)	30s	RED → page	P1 (PagerDuty)
MON-002	Throughput (Records/Sec)	Detect stalls / collapse in processing rate per domain	Prometheus	rate(records_processed_total[5m]) by (domain, operation)	Time-series (multi-line by domain)	15s	Warning if < 50 % of baseline for the domain	P3 Warning
MON-003	Latency Distribution	Track end-to-end and API latency; catch tail-latency regressions	Prometheus	histogram_quantile(0.95, rate(api_duration_seconds_bucket[5m]))	Time-series (P50/P95/P99 overlay)	15s	P95 > 5s	P3 Warning
MON-004	Error Rate (%)	Proportion of failed operations, split by error class	Prometheus	rate(errors_total[5m]) / rate(requests_total[5m]) * 100 by (error_class)	Time-series + threshold band	15s	> 5 %	P1 Critical
MON-005	DLQ Depth	Backlog of unprocessable messages in the dlq topic; leading indicator of systemic data-quality or load failure	Prometheus (Kafka exporter)	kafka_consumer_group_lag{topic="dlq"} + 24 h trend line	Time-series + current-value stat	30s	> 500 messages	P1 Critical
MON-006	Reconciliation Status	Last reconciliation verdict per domain with variance amount	Prometheus (Reconciliation Service metrics)	Last recon result per domain: RECONCILED / VARIANCE / BREAK with variance amount (recon_variance_amount gauge)	Table (per-domain rows, coloured status)	5m	Any BREAK	P2 High
MON-007	SAP System Health	Guard the RFC pool (≤ 50 concurrent) and SAP responsiveness; protect dialog users	Prometheus (SAP exporter)	sap_rfc_pool_utilisation, sap_response_time_ms, sap_work_process_available	Gauge cluster + time-series	30s	RFC pool > 90 %	P1 Critical
MON-008	FinSight API Health	Watch destination API latency, remaining rate-limit headroom, and token TTL	Prometheus (Loader metrics)	finsight_api_response_ms, finsight_rate_limit_remaining, finsight_token_expiry_seconds	Gauge cluster + time-series	30s	Rate-limit headroom < 10 %	P2 High
MON-009	Data Freshness	Age of the most recent successful load per domain against the ≤ 4 h SLA	Prometheus	time() - max(last_successful_load_timestamp) by (domain)	Bar gauge (per domain, SLA line)	60s	Any domain exceeds SLA (>4 h)	P3
MON-010	Resource Utilisation	Node-level CPU / memory / disk / network for the integration cluster	Prometheus (node_exporter)	node_cpu_utilisation, node_memory_utilisation, node_disk_utilisation, node_network_bytes	Time-series (per node)	15s	Any resource > 90 %	P2 High
MON-011	Kafka Consumer Lag	Per-topic / per-partition consumer lag — detects Transform or Loader falling behind producers	Prometheus (Kafka exporter)	kafka_consumer_group_lag by (topic, partition)	Heatmap / time-series by topic	15s	> 50 000 msgs	P2 High
MON-012	Circuit Breaker Status	State of every inter-service breaker (SAP RFC, FinSight API, per domain)	Prometheus	circuit_breaker_state{service} — 0=Closed, 1=Open, 2=Half-Open	State-timeline / status grid	5s	Any breaker Open	P2